02 - Text Preprocessing

Objective

The objective is to clean the `designation` and `description` fields before modelling. The same fixed rules are applied to the labelled training catalogue and the official test catalogue.

Report 1 found missing descriptions, encoded HTML characters and HTML tags. The cleaning removes this formatting noise while preserving product names, accents, numbers, units and model codes. No feature extraction or model training is performed here.

1. Load the Data

The labelled training catalogue and official test catalogue are loaded separately. Each product keeps its original `row_id`.

	catalogue	products
0	labelled training	84916
1	official test	13812

2. Clean the Text

For each catalogue, missing descriptions become empty text, HTML formatting is removed, uppercase is converted to lowercase, repeated spaces are removed, and the cleaned designation and description are combined.

```
Both catalogues were cleaned using the same rules.
```

3. Check Before and After Examples

The code automatically finds two short descriptions with HTML tags and two with encoded characters, then displays the original and cleaned versions.

	before	after
64477	Volet de skimmer	volet de skimmer
83375	Raccord en T 1/4	raccord en t 1/4
21098	pere noel mamas & papas	pere noel mamas & papas
48782	coffret de l'écrivain	coffret de l'écrivain

4. Check Vocabulary Cleaning

Report 1 found HTML words such as `br`, `strong`, `ul` and `li` in the raw vocabulary. Their occurrences are counted before and after cleaning to verify that formatting terms are removed.

	raw occurrences	cleaned occurrences
br	124213	561
strong	50399	303
ul	14724	85
li	109355	1037

5. Compare Top Words Before and After Cleaning

Figure 8 of Report 1 showed the top TF-IDF words per class on the raw text, and HTML terms such as `br`, `li` and `strong` were among them. The same view is repeated here on the raw and the cleaned text.

Report 1 used only French stopwords, but classes 10 and 40 contain English text and class 10 also contains German. This view therefore removes French, English and common German stopwords. The HTML terms that Report 1 added to its list are not included, so the cleaning alone must remove them. This view is only for inspection and removes nothing from the data.

prdtypecode	raw text	cleaned text
10	tome, edition, guide, histoire, book, livre, vie	tome, edition, guide, histoire, book, livre, édition
40	import, br, titres, ni, japonais, jeu, edition	import, titres, ni, japonais, jeu, edition, sous
50	br, nintendo, ps4, manette, console, jeu, xbox	nintendo, ps4, manette, console, jeu, xbox, wii
60	br, console, oui, jeu, ecran, tactile, communication	console, oui, jeu, ecran, communication, tactile, affichage
1140	figurine, pop, star, wars, shirt, cm, funko	figurine, pop, star, wars, shirt, cm, funko
1160	carte, yu, gi, oh, vf, mtg, magic	carte, yu, gi, oh, vf, mtg, magic
1180	warhammer, heroclix, oop, masque, figurine, prince, 40000	warhammer, heroclix, oop, masque, figurine, prince, 40000
1280	39, enfants, doudou, peluche, br, jouet, jouets	enfants, doudou, peluche, jouet, jouets, puzzle, bébé
1281	jeu, enfants, cartes, 39, game, br, jouet	jeu, enfants, cartes, game, jouet, jouets, cards
1300	drone, rc, dji, générique, 39, batterie, mavic	drone, rc, dji, générique, batterie, mavic, quadcopter
1301	chaussettes, bébé, filles, garçons, longueur, billard, br	chaussettes, bébé, filles, garçons, longueur, billard, taille
1302	39, pêche, br, stress, est, vous, stream	pêche, est, stress, vous, stream, lampe, enfants
1320	bébé, br, 39, bavoir, sac, imprimé, cm	bébé, bavoir, sac, imprimé, cm, poussette, rose
1560	br, 39, li, cm, matelas, table, bois	cm, matelas, bois, table, est, vous, hauteur
1920	oreiller, taie, coussin, 39, canapé, taille, plaît	oreiller, taie, coussin, canapé, taille, plaît, vous
1940	bio, br, café, sucre, 39, chocolat, sachet	bio, café, sucre, chocolat, sachet, lait, capsules
2060	39, br, led, vous, noël, décoration, diamant	led, vous, noël, décoration, lumière, diamant, lampe
2220	br, chien, 39, pet, chat, animaux, compagnie	chien, pet, chat, animaux, compagnie, taille, animal
2280	01, france, journal, 03, revue, magazine, paris	01, france, journal, 03, revue, magazine, paris
2403	lot, livres, tomes, volumes, tome, revues, collection	lot, livres, tomes, volumes, tome, revues, collection
2462	jeux, xbox, lot, wii, playstation, jeu, ps3	jeux, xbox, lot, wii, playstation, jeu, ps3
2522	notes, a5, papier, format, carnet, cm, pages	notes, a5, papier, format, carnet, cm, mm
2582	li, br, 39, strong, cm, jardin, est	cm, jardin, est, table, dimensions, aluminium, hauteur
2583	piscine, li, 39, br, eau, pompe, strong	piscine, eau, pompe, filtration, cm, spa, bâche
2585	li, br, 39, strong, mm, aspirateur, vous	aspirateur, mm, vous, est, bâche, votre, tondeuse

prdtypecode	raw text	cleaned text
2705	eacute, qui, est, rsquo, son, agrave, qu	qui, est, son, qu, il, que, elle
2905	li, strong, br, 39, ul, div, jeu	jeu, windows, directx, vous, téléchargement, gb, ram

Observations

- On the raw text, 18 of the 27 classes have an HTML term (`br` , `li` , `strong` , `39` , `eacute`) in their top six words. On the cleaned text, none do.
- Real product words take their place. Class `2583` now shows `filtration` and `spa` , and class `2905` , whose raw top words were all HTML, now shows `jeu` , `windows` and `directx` .
- With English and German stopwords added, class 10 shows book words (`tome` , `edition` , `guide` , `histoire`) instead of `the` , `of` and `and` .
- Some common French words remain, such as `est` , `vous` and `qui` , because the French list is short. Stopwords are not removed from the modelling text, because TF-IDF already gives low weight to words that appear everywhere. A full multilingual stopword list can be tested later as a model option.

6. Check Repeated Text

Repeated cleaned titles are counted but not removed. Some may be legitimate catalogue entries, and some identical titles have different labels. This check uses only the labelled training catalogue.

	count
rows with repeated titles	4120
repeated-title groups	1443
groups with conflicting labels	157

7. Attach the Shared Split

`02_preprocessing.ipynb` saves one training/validation assignment for every labelled `row_id` . Text and image preprocessing both join this same file, so they evaluate the same products. The official test catalogue remains separate.

split	products
train	67932
validation	16984

8. Validate and Save

The checks confirm that no products were lost and every product has usable combined text. The original files remain unchanged.

	catalogue	products_saved
0	labelled training	84916
1	official test	13812

Conclusion

Both catalogues use the same fixed cleaning rules. The labelled text also carries the shared training/validation assignment used by the image workflow. Learned transformations such as TF-IDF must later be fitted on training rows only.